Draven Chen

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Professional Summary

Computer Science undergraduate with a track record of shipping complete, production-quality software: five public releases in two years, including a commercial developer tool and a configurable rules engine, built and maintained solo. Specializes in systems design, modular architecture, and editor tooling. Comfortable owning a feature end to end and explaining technical decisions to non-technical stakeholders.

Education

Southern New Hampshire University (SNHU)

Bachelor of Science, Computer Science. 31 skill credentials independently verified through Credly, including Python, Java, Algorithms, Software Architecture, Systems Design, Database Design and SQL. Additional coursework completed in Calculus I-III and Differential Equations. *Expected mid-2027*

Relevant Coursework: Data Structures & Algorithms, Systems Architecture, Object-Oriented Programming, Software Engineering Principles, Databases, Operating Systems

Projects

crimSun Power Engine

Configurable Rules Engine — TypeScript, Vite

- Designed a configurable rules engine that evaluates a contest between two entities and exposes its full reasoning, built as a shared backbone so logic changes propagate from one location rather than many.
- Implemented a multi-tier resource system with two-zone sigmoidal scaling, resolving both sides in a single pass.
- Surfaced every step of the computation in a live inspection interface so non-engineer stakeholders can verify results without reading code.
- Shipped as a self-contained browser application with no server dependency and no install required.

VERA AI

Static Analysis Tool — Godot/GDScript, sold commercially on itch.io

- Wrote a static analyser that validates finite state machine transitions at edit time and flags faults inline, catching logic errors before runtime.
- Architected the tool to run exclusively at edit time, adding zero overhead to the production build.
- Packaged, documented, and published it independently as a commercial product, including buyer support material.

Sunnyside World

2D Simulation Application — Unity, C#, WebGL

- Built the full application loop across resource management, crafting, time-of-day simulation, and persistent state across sessions.
- Separated subsystems (inventory, agriculture, time) into independent modules with clean interfaces so each could evolve without destabilising the others.
- Optimized the asset pipeline for WebGL export, keeping load times viable in a browser environment.

Tiny Swords Last Stand

Wave-based Strategy Application — Unity/Godot, WebGL

- Drove wave scheduling from a data configuration layer rather than hard-coded values, making balance adjustments a data edit with no code changes required.
- Implemented entity behaviour as a finite state machine (approach, attack, retreat) designed for cheap extension to new entity types.
- Wired the UI directly to application state to eliminate feedback lag and ensure display accuracy.

Dragon Ball Sol

Combat Prototype — Unity, C#, Source on GitHub

- Integrated the Power Engine rules system into a real-time combat prototype, validating that the engine functions correctly inside a live application.
- Wired live state values (power, resource levels, active modifiers) into an in-application HUD.

crimSun-chrome-mcp

Open Source Tool — TypeScript

- Implemented a Model Context Protocol integration for browser-based tool workflows, published openly on GitHub.

academic-assignments-project-portfolio

Open Source

- Published coursework and technical assignments from the Computer Science degree openly on GitHub.

Skills

Languages: C#, C++, Python, TypeScript, Java, JavaScript, SQL, GDScript, HTML

Engineering: Systems architecture, Object-oriented design, Finite state machines, Modular software design, Static analysis, Algorithmic optimisation, Memory management, WebGL build optimisation, REST APIs, Database design

Tools & Engines: Git, GitHub, Unity, Godot, Unreal Engine (C++ and Blueprints), Vite

Practice: Competitive programming on LeetCode, Codewars, and CodinGame (C++, Java, Python)

AUDIENCE & COMMUNICATION

- Operate two YouTube channels covering technical and competitive topics.
- Produced a video credited in a platform's published history with growing its daily active users from ~250 to 550-600 within two weeks and doubling its subscriber base.
- Ranked first on two competitive leaderboards with a 14:1 win-to-loss ratio across ~3,300 recorded matches.